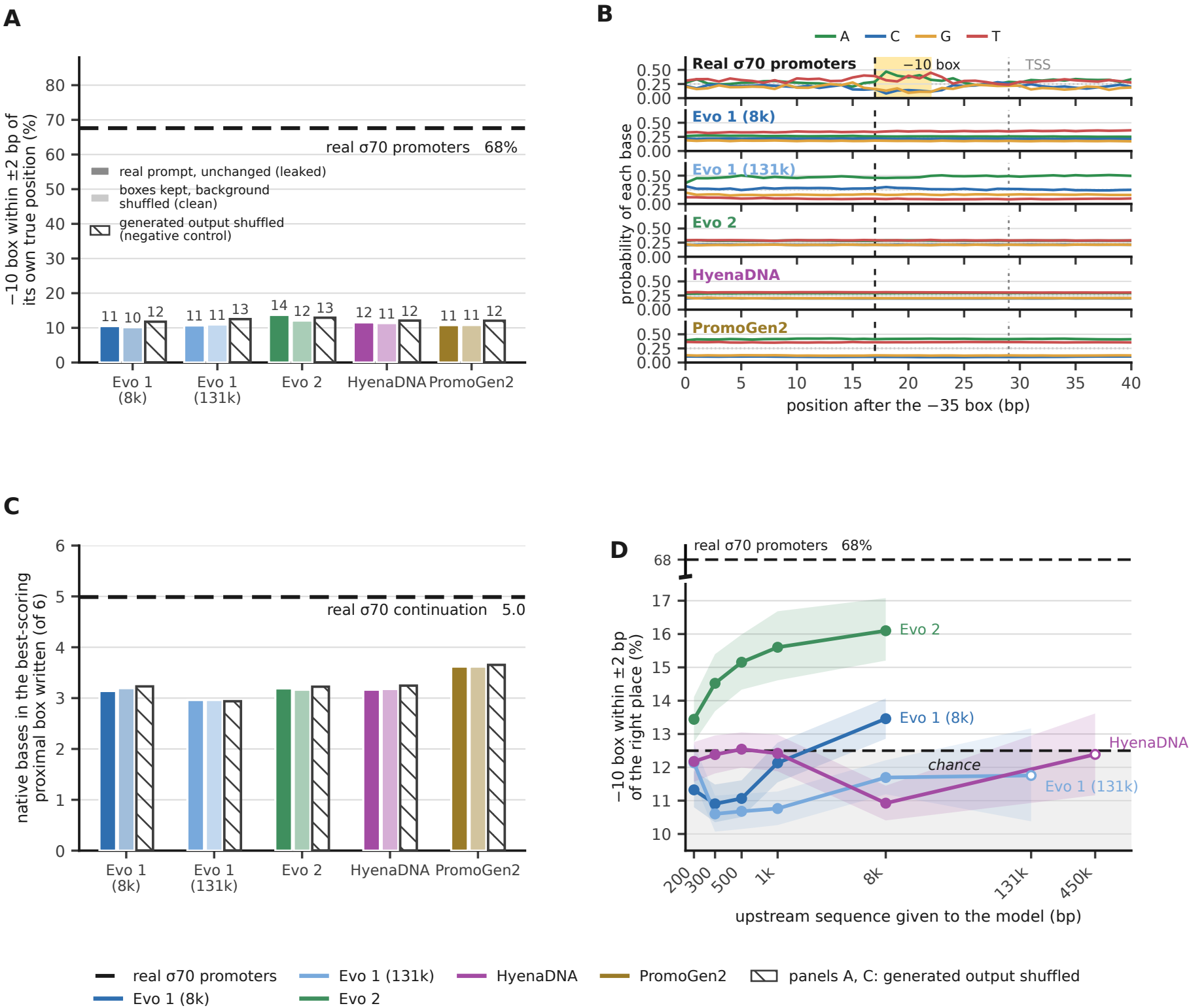
A
−10 box within ±2 bp of its own true position (%)
real σ70 promoters 68%
real prompt, unchanged (leaked)
boxes kept, background shuffled (clean)
generated output shuffled (negative control)
11 10 12
11 11 13
14 12 13
12 11 12
11 11 12
Evo 1 (8k)
Evo 1 (131k)
Evo 2
HyenaDNA
PromoGen2
B
A C G T
Real σ70 promoters
−10 box
TSS
Evo 1 (8k)
Evo 1 (131k)
Evo 2
HyenaDNA
PromoGen2
probability of each base
position after the −35 box (bp)
C
native bases in the best-scoring proximal box written (of 6)
real σ70 continuation 5.0
D
real σ70 promoters 68%
−10 box within ±2 bp of the right place (%)
Evo 2
Evo 1 (8k)
HyenaDNA
chance
Evo 1 (131k)
upstream sequence given to the model (bp)
real σ70 promoters
Evo 1 (8k)
Evo 1 (131k)
Evo 2
HyenaDNA
PromoGen2
panels A, C: generated output shuffled